

JavaScript Complete Road Map

Don't overwhelm to learn JavaScript, JavaScript is only this much

1. Variables

- var
- let
- const

2. Data Types

- number
- string
- boolean
- null
- undefined
- symbol

3. Declaring variables

- var
- let
- const

4. Expressions

Primary expressions

- this
- Literals
 - []
 - {}
- function
- class
- function*
- async function
- async function*
- /ab+c/i
- string
- ()

Left-hand-side expressions

- Property accessors
 - ?.
- new
- new .target
- import.meta
- super
- import()

5. operators

- Arithmetic Operators: +, -, *, /, %

- Comparison Operators: ==, ===, !=, !==, <, >, <=, >=
- Logical Operators: &&, ||, !

6.Control Structures

- if
- else if
- else
- switch
- case
- default

7.Iterations/Loop

- do...while
- for
- for...in
- for...of
- for await...of
- while

8.Functions

- Arrow Functions
- Default parameters
- Rest parameters
- arguments
- Method definitions
- getter
- setter

9.Objects and Arrays

- Object Literal: { key: value }
- Array Literal: [element1, element2, ...]
- Object Methods and Properties
- Array Methods: push(), pop(), shift(), unshift(), splice(), slice(), forEach(), map(), filter()

10.Classes and Prototypes

- Class Declaration
- Constructor Functions
- Prototypal Inheritance
- extends keyword
- super keyword
- Private class features
- Public class fields
- static
- Static initialization blocks

11.Error Handling

- try,
- catch,
- finally (exception handling)

ADVANCED CONCEPTS:-

12.Closures

- Lexical Scope
- Function Scope
- Closure Use Cases

13.Asynchronous JavaScript

- Callback Functions
- Promises
- async/await Syntax
- Fetch API
- XMLHttpRequest

14.Modules

- import and export Statements (ES6 Modules)
- CommonJS Modules (require, module.exports)

15.Event Handling

- Event Listeners
- Event Object
- Bubbling and Capturing

16.DOM Manipulation

- Selecting DOM Elements
- Modifying Element Properties
- Creating and Appending Elements

17.Regular Expressions

- Pattern Matching
- RegExp Methods: test(), exec(), match(), replace()

18.Browser APIs

- localStorage and sessionStorage
- navigator Object
- Geolocation API
- Canvas API

19.Web APIs

- setTimeout(), setInterval()
- XMLHttpRequest
- Fetch API
- WebSockets

20.Functional Programming

- Higher-Order Functions
- map(), reduce(), filter()
- Pure Functions and Immutability

21.Promises and Asynchronous Patterns

- Promise Chaining
- Error Handling with Promises
- Async/Await

22.ES6+ Features

- Template Literals
- Destructuring Assignment
- Rest and Spread Operators
- Arrow Functions
- Classes and Inheritance
- Default Parameters
- let, const Block Scoping

23.Browser Object Model (BOM)

- window Object
- history Object
- location Object
- navigator Object

24.Node.js Specific Concepts

- require()
- Node.js Modules (module.exports)
- File System Module (fs)
- npm (Node Package Manager)

25.Testing Frameworks

- Jasmine
- Mocha
- Jest

React ❤️ for more

YouTube Playlists:-

1.FREECODE CAMP

<https://youtu.be/PkZNo7MFNFg?si=ngyDRgeaBsEqPDkF>

2. PRASHANTH SIR (HINDI LANGUAGE)

<https://youtu.be/cpoXLj24BDY?si=4L1ugEjq2K07mh5x>

3.BRO CODE

<https://youtu.be/lfmq-EJ8gm4?si=RUuZ8gCUQ2MxJBXA>

4.SURESH TECHS(TELUGU LANGUAGE)

<https://youtu.be/n87Hns72pBM?si=FRJJ1yADcUCwFaMu>

5.All 100+ CONCEPTS EXPLAINED IN ONE VIDEO (*****)

https://youtu.be/lkIFF4maKMU?si=yoFO_4dWko6la8Pg

Git Project Resources:-

- 1.<https://github.com/CodeTraversal/JavaScript-Projects/tags>
- 2.<https://github.com/solygambas/html-css-javascript-projects>
- 3.<https://github.com/ZeroOctave/ZeroOctave-Javascript-Projects>
- 4.<https://github.com/zero-to-mastery/javascript20-projects>

Projects Ideas

1. Tip calculator ✓
2. Cost split ✓
3. Calculator
4. Form validator
5. Color picker ✓
6. Flyout navigation
7. Hover effect ✓
8. Slide-in/out
9. Popup
10. Tooltip ✓
11. JS animations
12. To-do list
13. Quiz
14. Astrology prediction
15. Color-scale generator
16. E-commerce - color change of product
17. Chat
18. Conversion of units
19. gif search

- 20. Tweet search
- 21. Random image
- 22. Quote generator
- 23. Sliders
- 24. Tabs
- 25. Progress bar
- 26. Word count ☒
- 27. Search for word in a page
- 28. Accordion

THE END